# Sub-sampling PLL with feedforward noise cancellation

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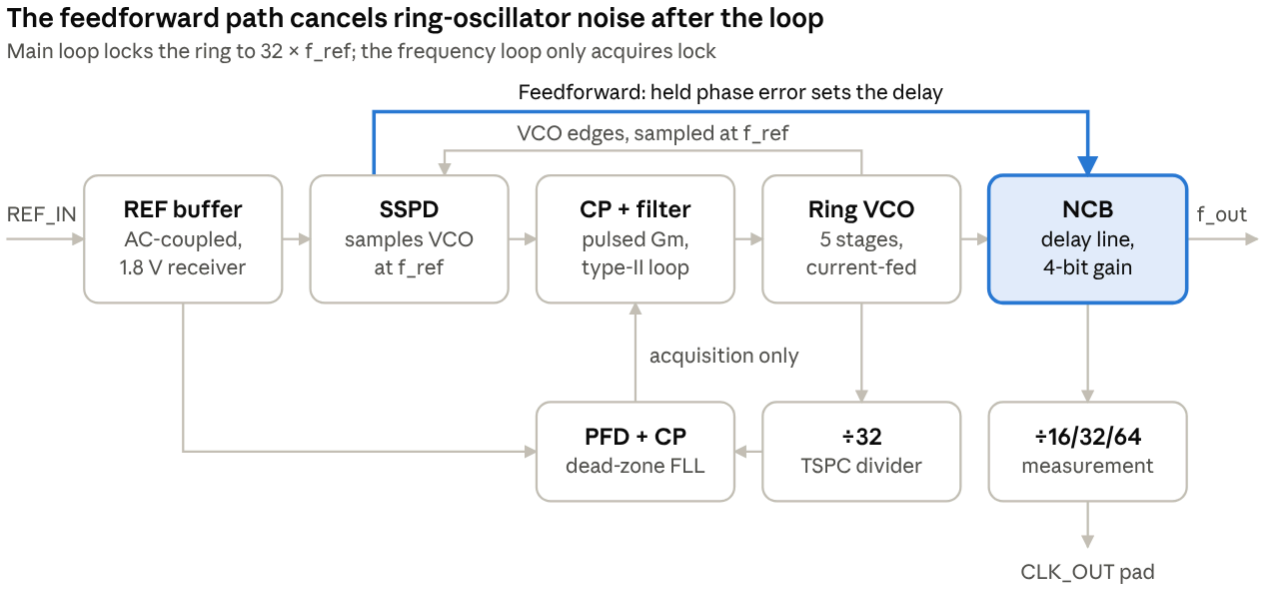
A 1.6 GHz ring-oscillator PLL for SKY130 that multiplies a 50 MHz crystal reference by 32 and divides the result into 800, 400, 200 and 100 MHz clocks. A sub-sampling phase detector keeps detector noise out of the $N^2$ multiplication, and feedforward noise cancellation removes about a third of the ring oscillator's jitter after the loop. Target: 1.6 ps rms jitter (10 kHz–40 MHz) at about 7.4 mW in under 0.1 $mm^2$, verified post-layout with open-source tools.

## 1. Type of IP block

An integer-N clock-multiplying PLL (phase-locked loop) for system-on-chip clocking, with a fixed multiplication ratio N = 32. An external 45–55 MHz crystal oscillator gives 1.44–1.76 GHz. The nominal 50 MHz reference gives 1.600 GHz, which on-chip binary dividers turn into 800, 400, 200 and 100 MHz. The architecture is a ring-oscillator sub-sampling PLL with feedforward noise cancellation, after Nagam and Kinget (IEEE JSSC, 2018), built from 1.8 V core devices with a 3.3 V supply for the oscillator current feed.

The first use is a low-jitter sampling clock for data converters: 1.6 ps rms keeps a 100 MHz input at about 60 dB SNR, roughly 10 bits. The divided outputs also give SKY130 digital logic and I/O clean 100–400 MHz clocks. No PLL was among the 25 blocks taped out in Chipalooza Challenge #1, whose oscillators are all low-frequency, and this block fills that gap.

# 2. Functional description




Block diagram · main loop, feedforward cancellation, frequency acquisition, measurement output

The loop locks a 5-stage pseudo-differential ring oscillator to 32 × f_ref. The noise-cancellation block then removes most of the ring's remaining jitter before the clock leaves the block.

- **Sub-sampling phase detector (SSPD).** It samples the 1.6 GHz oscillator output directly at the reference rate, and the held voltage is the phase error. The main loop has no feedback divider, so phase-detector and charge-pump noise are not multiplied by $N^2$.
- **Charge pump and loop filter.** A pulsed-Gm charge pump and a type-II RC filter set the fine tuning voltage, with a loop bandwidth of about 2.5 MHz.
- **Ring oscillator.** Five pseudo-differential stages. The odd stage count makes the common-mode loop inverting, so the ring cannot latch at power-up. A 5-bit current DAC feeds the ring from 3.3 V and sets the coarse frequency, and its range keeps the ring's supply node below 1.8 V at every corner.
- **Feedforward noise cancellation (NCB).** The held phase error also drives a voltage-controlled delay line after the oscillator. With its gain set to the inverse of the detector gain, the delay subtracts the oscillator's own phase deviation, up to about f_ref/8 (6 MHz). A 4-bit code sets the gain and is calibrated in the foreground with a test tone. With the block disabled, the PLL works normally without cancellation.
- **Frequency acquisition.** A sub-sampling detector locks to whichever harmonic is nearest, so a dead-zone frequency-locked loop (÷32 divider, PFD, charge pump) pulls the oscillator to 32 × f_ref. It goes quiet once the phase error is inside its dead zone, and catches any slip to a neighbouring harmonic.

- **Outputs.** The full-rate clock and its ÷2, ÷4, ÷8 and ÷16 taps (800, 400, 200 and 100 MHz at a 50 MHz reference) go to the system-on-chip. A ÷16, ÷32 or ÷64 copy goes off-chip on CLK_OUT for measurement.

## 3. I/O list, including test ports

On the Chipalooza harness the block uses all three dedicated pads, all four shared analog lines (DC and low-frequency signals only), one bandgap-referenced current, 19 of the 24 digital controls and 4 of the 12 digital test outputs.

| Signal | Type | Connection | Function |
|---|---|---|---|
| REF_IN | Analog input | Dedicated pad 1, analog pad | 45–55 MHz reference from an external crystal oscillator. The pad self-biases at VDDA/2 and feeds an AC-coupled on-chip receiver, so the pad stays within 0 V to VDDA and a 3.3 V swing never reaches the 1.8 V core. Needs a direct, low-resistance path to the pad |
| CLK_OUT | Digital output | Dedicated pad 2 | Measurement clock: f_out/16 (100 MHz nominal), f_out/32 (the frequency-loop divider output, 50 MHz) or f_out/64. Selected over SPI |
| VTEST | Analog input/output | Dedicated pad 3, analog pad | Injects a test tone into the oscillator current feed, for noise-cancellation calibration and supply-noise tests. In monitor mode it reads the oscillator supply node instead |
| AMON[3:0] | Analog outputs | Shared analog lines 0–3 | Buffered DC monitors: tuning voltage, oscillator supply node, held phase-detector output, bias test current |
| VDDA | Power | Slot 3.3 V | Oscillator current feed (CCO) and pad interface |

| Signal | Type | Connection | Function |
|---|---|---|---|
| VCCD | Power | Slot 1.8 V | Core: phase detector, charge pump, loop filter, ring oscillator, dividers, noise-cancellation block |
| VSS | Ground | Slot ground | |
| IBIAS | Bias input | 1 bandgap-referenced current source | Reference for charge pump, phase-detector buffers and current DAC |

**Digital controls (19 of 24, SPI):** pll_en, fll_en, ffnc_en, cco_code[4:0] (coarse frequency), ncb_code[3:0] (cancellation gain), icp_trim[1:0], clkout_sel[1:0], clkout_en, vtest_mode, amon_en. The remaining five are reserved.

**Digital test outputs (4 of 12):** lock_det, fll_active, fll_dir (too fast or too slow), cal_out (sign of the calibration error).

**Core-level outputs** for reuse in other chips: CLK_FULL (f_out) and CLK_DIV2 to CLK_DIV16. They are not brought out on the harness. The bandgap-referenced bias voltage is not used.

## 4. Target specification

Minimum and maximum values hold across the full operating range below, verified post-layout. Typical values are for the tt corner at 27 °C, nominal supplies, a 50 MHz reference and calibrated codes. Unless noted, performance is at the oscillator output with noise cancellation (FFNC) on.

**Operating conditions**

| Parameter | Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VDDA | | 2.97 | 3.3 | 3.63 | V |
| VCCD | | 1.62 | 1.8 | 1.98 | V |
| Temperature | Junction | −40 | 27 | 110 | °C |
| Reference frequency f_ref | External crystal oscillator | 45 | 50 | 55 | MHz |
| Reference amplitude | Sine or square, AC-coupled off-chip; or a 0 V to VDDA square wave | 0.4 | 1.0 | 3.3 | V pp |
| Reference phase-noise floor | ≥ 100 kHz offset; condition for the jitter and phase-noise rows | | | −155 | dBc/Hz |

## Performance

| Parameter | Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Output frequency f_out | 32 × f_ref | 1.440 | 1.600 | 1.760 | GHz |
| Divided outputs | f_out/2, /4, /8, /16 at f_ref = 50 MHz | | 800, 400, 200, 100 | | MHz |
| CLK_OUT frequency | /16 setting | 90 | 100 | 110 | MHz |
| RMS jitter, FFNC off | Integrated 10 kHz–40 MHz | | 2.4 | 4.7 | ps |
| RMS jitter, FFNC on | Integrated 10 kHz–40 MHz | | 1.6 | 3.2 | ps |
| Jitter reduction by FFNC | After gain calibration | 20 | 35 | | % |
| Phase noise, FFNC off | 300 kHz offset | | −111 | −102 | dBc/Hz |
| Phase noise, FFNC on | 300 kHz offset | | −120 | −110 | dBc/Hz |
| Reference spur, FFNC off | At f_out ± f_ref | | −55 | −45 | dBc |
| Reference spur, FFNC on | At f_out ± f_ref | | −50 | −42 | dBc |
| Lock time | From pll_en with coarse code set, until the phase error stays within ±10 ps | | 5 | 20 | µs |
| Power, FFNC on | PLL core; excludes output dividers and CLK_OUT pad driver | | 7.4 | 12 | mW |
| Power, FFNC off | Noise-cancellation block disabled | | 6.6 | 11 | mW |
| Power, output dividers | All four taps (/2 to /16) enabled | | 0.5 | 1.0 | mW |
| Jitter figure of merit | $10 \cdot \log10[(\sigma/1\ \mathrm{s})^2 \cdot (P/1\ \mathrm{mW})]$, FFNC on | | −227.4 | | dB |

| Parameter | Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Area | Standalone layout, excluding pads | | 0.06 | 0.10 | mm² |

**Absolute limits** (exceeding these may damage the block; operation is only guaranteed within the operating conditions)

| Parameter | Min | Max | Unit |
|---|---|---|---|
| VDDA | 0 | 3.63 | V |
| VCCD | 0 | 1.98 | V |
| REF_IN, VTEST pin voltage | −0.3 | VDDA + 0.3 | V |
| Internal oscillator supply node, any cco_code | | 1.80 | V |
| Junction temperature | −40 | 125 | °C |

The internal oscillator supply limit is a design constraint, not a user setting: the current DAC range and an on-chip clamp keep the 1.8 V ring devices within rating at every code and corner.

## 5. Test plan outline

Every measurement uses the one CLK_OUT pin, so cancellation on and off are compared with the same cable, divider and instrument. The 1.6 GHz clock never leaves the chip.

1. **Bring-up.** Apply supplies and bias, check SPI access, and read the slot current with each enable bit set in turn to get per-block power.
2. **Coarse calibration.** With the frequency loop enabled, sweep cco_code and record the codes where lock_det goes high. Repeat at each test temperature.
3. **Frequency and lock.** Measure CLK_OUT on a frequency counter: 100.000 MHz at f_ref = 50 MHz. For lock time, set CLK_OUT to /32, trigger an oscilloscope on the SPI write that sets pll_en, and time how long CLK_OUT takes to settle to a fixed phase against REF_IN. The status bits are read over SPI, so lock_det confirms lock but cannot time it. Repeat at f_ref = 45 and 55 MHz to confirm the range.
4. **Phase noise and jitter.** Measure CLK_OUT /16 on a phase-noise analyzer, or a spectrum analyzer with a phase-noise option. Add 20·log10(16) = 24.1 dB to refer phase noise to the oscillator output. Report rms jitter integrated from 10 kHz to 40 MHz directly: absolute jitter in seconds passes through an ideal divider unchanged. Repeat with ffnc_en off and on.

5. **Cancellation calibration.** Inject a 50 kHz tone on VTEST and step ncb_code from 0 to 15. Keep the code with the smallest tone sideband at CLK_OUT, then confirm the jitter reduction against step 4.
6. **Reference spur.** With N = 32, the spur sidebands on the 100 MHz output fall at 50 and 150 MHz, on top of the reference frequency and its third harmonic. Measure both lines in lock, then again with pll_en off and the reference still running, which shows the reference feedthrough alone. A true spur is a phase-modulation sideband, so the two lines are equal in level and stand well above the feedthrough. Add 24.1 dB to refer the spur to the oscillator output.
7. **Supply-noise sensitivity.** Inject 1 mVpp at 100 kHz on VTEST and compare the resulting spur with cancellation on and off.
8. **Temperature.** Repeat steps 3, 4 and 6 at the coldest and hottest temperatures the available equipment allows, up to −40 and 110 °C.

**Equipment needed:** a 50 MHz crystal oscillator with a phase-noise floor of −155 dBc/Hz or lower, a signal generator covering 45–55 MHz for the range test, a phase-noise analyzer or spectrum analyzer to at least 200 MHz, a 1 GHz oscilloscope, a function generator for the test tone, a frequency counter, a source-measure unit or bench supplies with current readout, and a SPI host. Measurement will take place at an external lab facility, listed with its equipment in the separate attachment.

## 6. Verification with open-source tools

Every result can be reproduced from the public repository (Apache 2.0) with xschem, ngspice, Magic, netgen, KLayout and CACE, inside the IIC-OSIC-TOOLS container. Commercial tools are used only as an optional cross-check and are never needed to rebuild or verify the block.

- **Corners:** tt, ss, ff, sf and fs; VDDA and VCCD at ±10%; −40, 27 and 110 °C; reference at 45, 50 and 55 MHz; plus Monte Carlo mismatch on the phase detector, charge pump and noise-cancellation block.
- **Jitter:** ngspice has no periodic-noise analysis, so jitter is built from parts that it can simulate. Oscillator phase noise comes from its impulse sensitivity function, extracted with charge-injection transients. Phase-detector, charge-pump, reference-buffer and cancellation-block noise come from ngspice noise analysis. A Python phase-domain model in the repository combines them. A closed-loop transient of the extracted netlist, with injected noise sources and edge timing extracted in Python, checks the combined result.
- **Sign-off:** Magic and KLayout DRC, netgen LVS and Magic RC extraction. The final DRC and LVS reports are committed to the repository.